

James Chen

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EDUCATION

Columbia University

M.S. in Computer Science

Dean's MS Academic Excellence Fellowship - \$15,000

Sep 2026 – Dec 2027

New York, NY

University of Massachusetts Lowell

B.S./M.S. in Computer Science; GPA: 4.0/4.0

Chancellor's Medals for Distinguished Academic Achievement

Jan 2024 – Dec 2025

Lowell, MA

WORK EXPERIENCE

Software Engineering Intern

UMass Lowell - Miner School of Computer and Information Sciences

May 2025 – Dec 2025

Lowell, MA

- Migrated 28,000+ Alliance to End Plastic Waste (AEPW) records into an analytical PostgreSQL schema, reducing query times by approximately 70% and strengthening data integrity.
- Designed and implemented AUDA, a Python analytics platform with a Typer CLI to automate data preprocessing, model execution, and visualization for reproducible experiments.
- Developed and evaluated machine learning workflows to detect anomalies, forecast trends, and reconstruct missing plastic waste and resin consumption data across 192 countries.
- Presented research progress and proposed approaches to a 12-member research group, explaining analytical limitations.

Backend Developer Intern

Take 2 Health

Mar 2021 – Sep 2021

Shenzhen, China

- Delivered a backend for a blood specimen management platform using Midway.js, with JWT authentication and RESTful APIs for specimen registration and processing status updates.
- Migrated 90+ API endpoints from Spring Boot to JavaScript/TypeScript, documented APIs in Swagger, and coordinated integration with frontend engineers.
- Designed a microservices architecture to isolate service failures and simplify maintenance, implementing an initial user service with gRPC communication.

Backend Developer

Juyingtong

Nov 2018 – Feb 2020

Zhuhai, China

- Developed a conference management backend for a WeChat mini-program with ThinkPHP 5, MySQL, Redis session authentication, and RESTful APIs.
- Implemented QR code check-in with WeChat geolocation validation, restricting attendance registration to users within the venue's allowed radius and one check-in per user per conference.
- Integrated WeChat Pay for conference registration payments and refunds upon cancellation.
- Deployed the application on Alibaba Cloud Linux servers, configured Nginx reverse proxy and load balancing, and managed application processes with systemd.

PROJECTS

- **Bloxtopia (bloxtopia.gg)**: Built a rewards platform for Roblox players with Next.js and Nodemailer, enabling users to complete sponsored tasks and redeem earnings for Robux. Collaborated with teammates to resolve frontend-backend integration issues by introducing a gateway layer that centralized mock data for testing and routed production requests to backend services.
- **Nemotron (github.com/DMax1314/Nemotron)**: Wrote fine-tuning scripts for a three-person team, coordinating execution on teammates' H200 GPU resources to improve NVIDIA Nemotron's reasoning accuracy from 30% to 85%.
- **Burrow (github.com/TypingHare/burrow)**: Created an extensible CLI platform with startup times under 20 milliseconds and built-in modules for configuration, HTTP server startup, and self-updates.

SKILLS

- **Programming Languages**: Java (4 yrs), Javascript/Typescript (4 yrs), Python (3 yrs), Go, C++, Kotlin, PHP, Lua, Rust, Bash, SQL, HTML/CSS
- **Frameworks and Libraries**: React, Spring Boot, Node.js, Vite, MUI, NestJS, Drizzle ORM, JUnit, Numpy, Pandas, Matplotlib, Scikit-learn, PyTorch, Typer
- **Databases and APIs**: PostgreSQL, MySQL, MongoDB, Redis, HTTPS, RESTful, GraphQL, WebSocket, Swagger
- **Tools and Platforms**: Git, GitHub, Docker, Nginx, systemd, AWS EC2/Lightsail, POSIX, npm/bun, Neovim
- **AI**: OpenAI API, Kaggle, NLP, Neural Networks, Transformer/GPT, LoRA, RLHF, PPO